

Comparison of CNN Model Performance for Fruit Freshness Classification Using Image Preprocessing Techniques on EfficientNetB0 and ResNet50V2

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Abstract

Automated quality inspection systems are crucial in the modern agricultural and food industries to ensure food safety, maintain market value, and minimize waste. This study presents a comparative evaluation of two Convolutional Neural Network (CNN) transfer learning architectures, namely EfficientNetB0 and ResNet50V2, for binary fruit freshness classification (fresh vs. rotten). To evaluate the impact of image enhancement, we compared a baseline scenario (resizing and normalization only) against a preprocessed scenario utilizing Gaussian Blur for noise reduction and Contrast Limited Adaptive Histogram Equalization (CLAHE) for contrast enhancement on the L channel of the LAB color space. The models were trained in two phases: head training with frozen backbones under strong data augmentation, followed by fine-tuning of top layers with light augmentation. Evaluation on the testing set was conducted using Test Time Augmentation (TTA) over seven steps. The experimental results demonstrate that the application of the preprocessing pipeline consistently improves performance across both models, increasing the F1-score of EfficientNetB0 from 0.8610 to 0.8889, and ResNet50V2 from 0.9161 to 0.9443. The ResNet50V2 model combined with preprocessing emerged as the optimal configuration, achieving the highest accuracy of 0.9444 and an F1-score of 0.9443, while misclassifying zero fresh images. These findings indicate that combining appropriate CNN architectures with specialized preprocessing techniques significantly enhances model sensitivity to subtle, localized surface spoilage, offering a robust solution for automated agricultural quality control.

Keywords— Fruit Freshness Classification, Convolutional Neural Networks, Deep Learning, Image Preprocessing, CLAHE, Transfer Learning.

1. Introduction

1.1. Background

Recent advances in Artificial Intelligence (AI) and Computer Vision have made significant contributions across various industries, especially agriculture and food production. One key application is the development of automated systems that can quickly assess fruit quality and freshness. Determining fruit freshness is crucial for food safety, maintaining market value,

and reducing waste. When rotten fruit is not detected early, it can quickly spoil surrounding produce, causing significant financial losses and potential health risks for consumers.

Traditionally, quality inspection is done manually by human workers. However, this approach has several drawbacks: it is time-consuming, inconsistent due to human fatigue, highly subjective, and sensitive to changing lighting conditions. To solve these problems, we need an automated system that is fast, consistent, and accurate. Deep learning, particularly Convolutional Neural Networks (CNNs), is highly suitable for this task because it can automatically learn and recognize visual patterns from image data.

This project evaluates and compares two popular transfer learning architectures: EfficientNetB0 and ResNet50V2. EfficientNetB0 is known for its efficiency and lightweight design. It uses a compound scaling method that balances network depth, width, and resolution to achieve high accuracy without requiring massive computational power. On the other hand, ResNet50V2 uses skip connections (residual connections) that allow gradients to flow directly through the network, preventing the vanishing gradient problem in deeper architectures.

To adapt these models for our specific task, the default classification heads were replaced with custom designs. EfficientNetB0 was equipped with a simple classifier consisting of Global Average Pooling, Batch Normalization, Dropout (50%), and a final classification layer. ResNet50V2 was equipped with a deeper classifier featuring Global Average Pooling, Batch Normalization, a Dense layer of 512 units with L_2 regularization, Dropout (50%), a Dense layer of 256 units with L_2 regularization, Dropout (30%), and the final classification layer.

However, a model's performance depends heavily on the quality of its input data. Therefore, image preprocessing is a critical step. In this study, we compare two dataset scenarios: a preprocessed dataset and an unprocessed baseline dataset. The preprocessing pipeline applies Gaussian Blur to reduce image noise, Contrast Limited Adaptive Histogram Equalization (CLAHE) to enhance contrast and make surface spots more visible, and pixel normalization. By testing all four model combinations (two architectures under two preprocessing scenarios), we aim to systematically analyze how preprocessing affects classification performance and find the best setup for classifying fresh and rotten fruit.

1.2. Problem Statement

Based on the background described above, this project addresses the following research questions:

1. How does a CNN-based model perform in classifying fruit freshness into fresh and rotten categories?
2. Do image preprocessing techniques (Gaussian Blur and CLAHE) significantly improve classification performance relative to an unprocessed baseline?

3. Which architecture (EfficientNetB0 or ResNet50V2) yields superior performance in fruit freshness classification?
4. To what extent does preprocessing influence accuracy, precision, recall, and F1-score for each model?
5. How capable are the models of detecting partially spoiled fruit whose visual characteristics closely resemble those of fresh fruit?

1.3. Objectives

The main objectives of this project are:

1. Build a binary fruit freshness classifier (fresh vs. rotten) using CNN-based transfer learning.
2. Compare model performance across two dataset scenarios: WITH preprocessing and WITHOUT preprocessing.
3. Analyze and quantify the effect of image preprocessing techniques on classification outcomes for each architecture.
4. Conduct a head-to-head comparison of EfficientNetB0 and ResNet50V2 on identical datasets.
5. Determine the best-performing model based on accuracy, precision, recall, and F1-score, evaluated using Test Time Augmentation (TTA).
6. Identify prediction error patterns through confusion matrix analysis across all four model variants.

1.4. Significance

The significance of this research lies in its practical application for the agricultural and food industries. By implementing an automated classification system, distributors and retailers can identify rotten fruit before it contaminates healthy produce, reducing post-harvest losses. Furthermore, this system ensures food safety for consumers by minimizing the risk of selling spoiled fruit, while simultaneously improving processing speeds and standardization compared to subjective, manual human inspections.

1.5. Scope and Limitations

To keep the project focused and clear, we establish the following boundaries:

- **Classification Scope:** The classification is strictly binary (fresh vs. rotten). The system does not identify the specific type of fruit.

- **Dataset Constraints:** The model is trained and evaluated using the publicly available Kaggle dataset "Fruits Quality (Fresh VS Rotten)".
- **Model Architectures:** We limit our deep learning models to EfficientNetB0 and ResNet50V2 using transfer learning.
- **Experimental Scenarios:** We evaluate two dataset conditions: a preprocessed scenario (using Gaussian Blur, CLAHE, and normalization) and a baseline scenario (using resizing and normalization only).
- **Evaluation Metrics:** Performance is measured using accuracy, precision, recall, F1-score, and confusion matrices, with all test predictions enhanced by Test Time Augmentation (TTA).
- **Implementation Context:** The project is implemented and analyzed within a computational notebook environment. Developing web/mobile applications or deploying to physical hardware (like cameras) is outside the scope of this work.

2. Related Work

Automated classification of agricultural commodity quality has been widely studied using deep learning. Mukhiddinov et al. [1] combined deep CNN feature extraction with an optimized YOLOv4 object detector to classify fresh and rotten fruits and vegetables. Their system was trained on a dataset of 12,000 images spanning 20 classes across five fruit and vegetable types, achieving 73.5% Average Precision (AP) with the improved YOLOv4—outperforming the standard YOLOv4 (72.6% AP). The architecture leverages a backbone for deep feature extraction and a neck module for capturing semantic and positional information, enabling robust classification under varying lighting and occlusion conditions. Their system targets practical deployment in food industry, supermarket, and assistive technology settings for visually impaired users. Building on deep residual architectures for fruit quality tasks, Shu et al. [2] developed a fruit freshness classification and detection model by integrating a Non-local Attention mechanism into the ResNet-101 network. The improved model outperforms conventional ResNet-101, VGG-16, and ResNet-50 across multiple evaluation metrics, with particularly strong robustness under complex real-world conditions such as varied lighting, cluttered backgrounds, and partial occlusion. The Non-local Attention mechanism enhances the model's capacity to detect subtle rotted areas and color discrepancies on fruit surfaces that standard convolutional layers tend to miss. Although the inclusion of Non-local Attention slightly increases inference time, performance remains within an acceptable range for real-time applications on edge devices. The study demonstrates the practical potential of attention-augmented deep networks for intelligent agriculture and smart logistics applications.

Beyond fruit-specific studies, transfer learning with established architectures provides a broader

foundation for model selection. Bobba [3] applied pre-trained ResNet50 and EfficientNet models to remote sensing image classification by fine-tuning them on the large-scale Eurosat dataset, achieving encouraging accuracy results. The study demonstrates that leveraging pre-trained models allows competitive performance even when labeled remote sensing data is limited, and evaluation using accuracy metrics and confusion matrices highlighted specific classes where the model encountered difficulty in differentiation. These transfer learning insights are broadly applicable to image classification tasks, including agricultural produce classification. On the data preparation side, Alomar et al. [4] surveyed data augmentation techniques used in image classification and segmentation, with a focus on addressing the overfitting problem common in CNN-based deep learning. In addition, they proposed a novel geometric augmentation technique called Random Local Rotation (RLR), which manipulates local spatial information within images without introducing any non-original pixel values. Experimental results demonstrated that RLR outperforms the traditional whole-image rotation technique in both classification and several segmentation tasks, confirming that targeted local augmentation strategies effectively complement existing augmentation pipelines and improve generalization.

3. Methodology

This project follows a step-by-step pipeline, starting from data loading to final model evaluation. The overall pipeline is illustrated in Fig. 1.

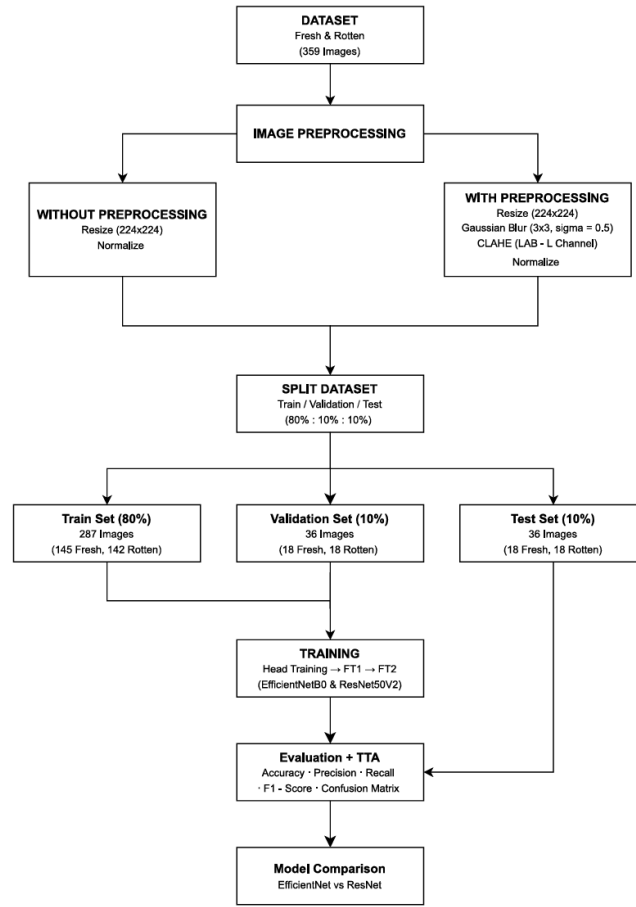


Figure 1: Pipeline of the project

3.1. Dataset

The dataset used in this project is "Fruits Quality: Fresh vs Rotten", created by Nour Abdoun and hosted on Kaggle [5]. This dataset was chosen because it has a balanced class distribution and contains a good variety of fruits, making it suitable for training a general binary classifier.

The dataset contains a total of 359 images: 181 images of fresh fruit and 178 images of rotten fruit. It includes many different types of fruit, such as strawberries, apples, bananas, kiwis, mangoes, grapes, cherries, pears, peaches, and oranges. Each image has a ground truth label representing whether the fruit is physically fresh or rotten. Although the dataset is relatively small, its balanced nature ensures that our evaluation metrics are fair and unbiased.

3.2. Dataset Splitting

To train and test the models properly, the dataset was split using stratified splitting into an 80:10:10 ratio. Stratification ensures that the ratio of fresh to rotten fruit remains the same in the training, validation, and testing sets:

- **Training Set:** 287 images (145 fresh, 142 rotten) — used to train the models.
- **Validation Set:** 36 images (18 fresh, 18 rotten) — used to tune parameters and prevent overfitting.
- **Testing Set:** 36 images (18 fresh, 18 rotten) — used for final evaluation on unseen data.

Additionally, we computed balanced class weights during training to handle any slight class imbalances.

3.3. Image Preprocessing

We designed two data scenarios to test the impact of preprocessing:

1. **WITHOUT Preprocessing (Baseline):** Images are resized to 224×224 pixels and normalized.
2. **WITH Preprocessing:** Images are resized, run through a preprocessing pipeline, and normalized:
 - **Gaussian Blur:** Applied using a 3×3 kernel with a standard deviation (σ) of 0.5. This smooths out random noise and small background textures, helping the model focus on the main fruit features.
 - **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Applied to the L channel in the LAB color space with a clip limit of 2.5 and grid size of 8×8 . CLAHE balances uneven lighting and enhances local contrast, making surface defects like spots, mold, or rot much clearer.
 - **Pixel Normalization:** Preprocessed images are normalized using the standard function of each respective architecture.

A comparison of images under both scenarios is shown in Fig. 2.

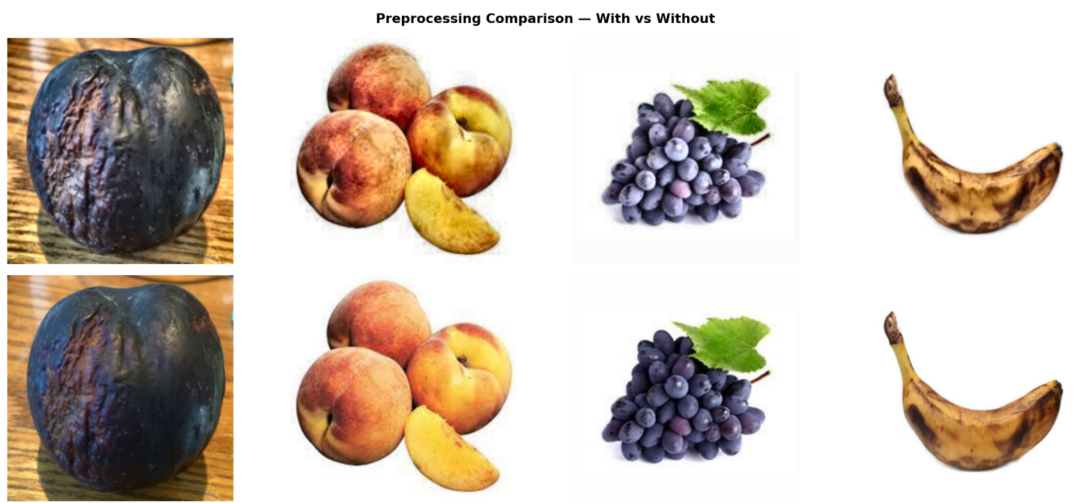


Figure 2: Comparison of images with and without preprocessing

3.4. Model Architectures

We utilized two architectures initialized with pre-trained ImageNet weights:

- **EfficientNetB0:** EfficientNetB0 is optimized for computational efficiency, balancing depth, width, and input resolution. It serves as our lightweight model. Its native normalization keeps pixel values in the $[0, 255]$ range, handling rescaling internally.
- **ResNet50V2:** ResNet50V2 uses skip connections to bypass layers, allowing gradients to flow directly during backpropagation and preventing the vanishing gradient problem. It has a larger capacity to learn detailed textures and color transitions. ResNet50V2 normalizes inputs to the $[-1, 1]$ range.

4. Implementation & Results

4.1. System Details

The system was implemented using the Keras and TensorFlow frameworks on a GPU-accelerated environment. The default classification tops of both networks were removed and replaced with custom heads designed to map features to the two target classes (fresh vs. rotten):

- **EfficientNetB0 Head:** Global Average Pooling 2D, Batch Normalization, Dropout (50%), and a final Dense layer with 2 units (softmax activation).
- **ResNet50V2 Head:** Global Average Pooling 2D, Batch Normalization, a Dense layer of 512 units (ReLU, $L_2 = 1 \times 10^{-4}$), Dropout (50%), a Dense layer of 256 units (ReLU, $L_2 = 1 \times 10^{-4}$), Dropout (30%), and a final Dense layer with 2 units (softmax activation).

4.2. Experiments

Model training was conducted using the Adam optimizer and Categorical Crossentropy loss (label smoothing = 0.1) in two distinct stages:

1. **Phase 1 (Head Training):** Backbone layers were frozen. The custom head was trained for up to 30 epochs at a learning rate of 1×10^{-3} with strong data augmentation (rotation up to 40° , vertical/horizontal flips, zoom up to 0.3, shear of 0.15, brightness range $[0.6, 1.4]$, channel shift of 30, and reflection fill mode).
2. **Phase 2 (Fine-Tuning):** A portion of the backbone layers was unfrozen (top 20 layers for EfficientNetB0, top 50 layers for ResNet50V2). The network was trained for up to 25 epochs at a lower learning rate of 5×10^{-5} under light data augmentation.

Training stability was maintained via `EarlyStopping` (patience of 10 epochs on validation accuracy) and `ReduceLROnPlateau` (patience of 5 epochs on validation loss, scaling factor of 0.4). During testing, Test Time Augmentation (TTA) was executed over 7 steps with light

augmentation parameters, and final probabilities were averaged.

4.3. Results Obtained

Table 1 summarizes the performance of all four model variants on the test set, evaluated using TTA.

Table 1: Evaluation Results of All Model Variants

Model Variant	Acc	Prec	Rec	F1
EfficientNet <i>with</i> Preprocessing	0.8889	0.8889	0.8889	0.8889
EfficientNet <i>without</i> Preprocessing	0.8611	0.8622	0.8611	0.8610
ResNet50V2 <i>with</i> Preprocessing	0.9444	0.9500	0.9444	0.9443
ResNet50V2 <i>without</i> Preprocessing	0.9167	0.9286	0.9167	0.9161

A visual comparison of these metrics is shown in Fig. 3, and the training curves are presented in Fig. 4.

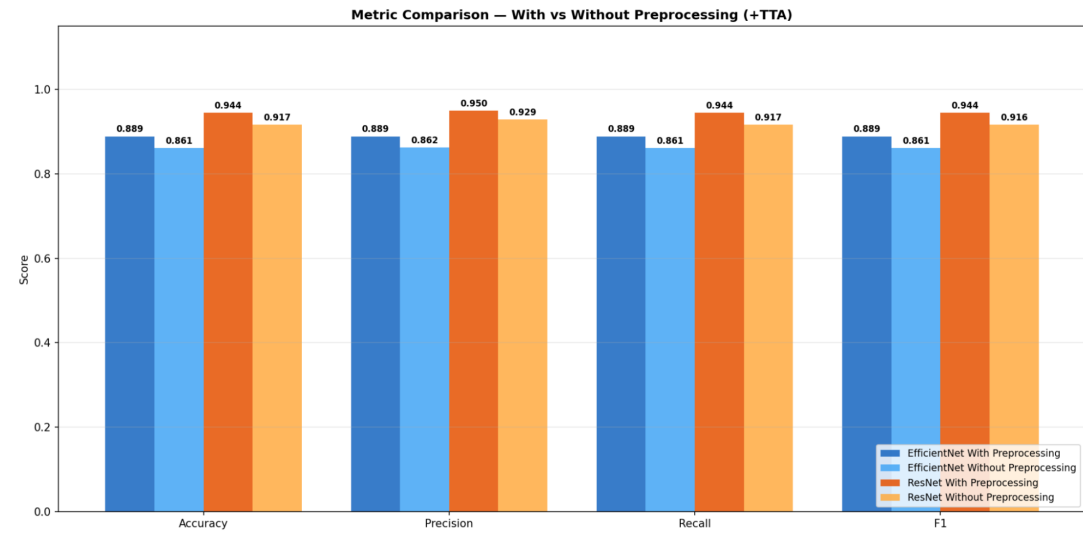


Figure 3: Comparison of accuracy, precision, recall, and F1-score across all trained models

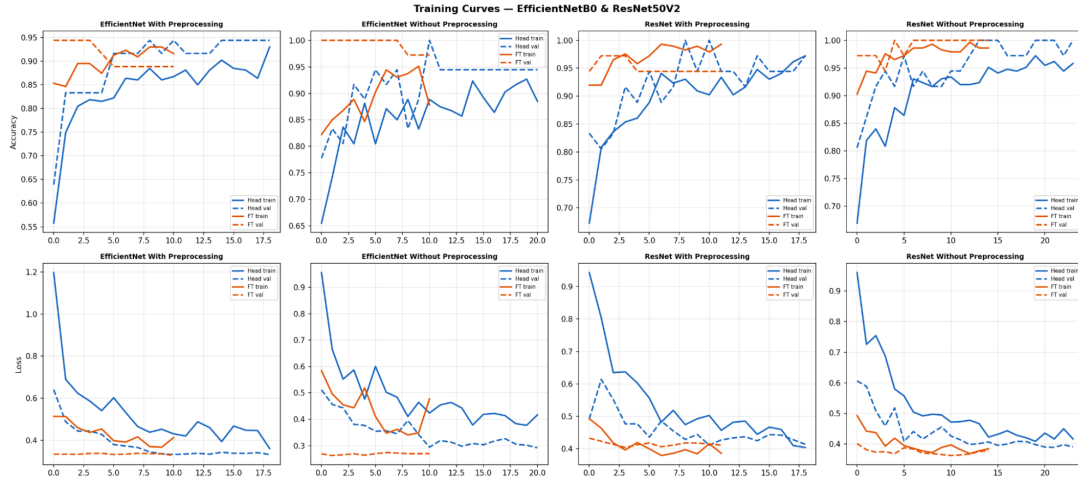


Figure 4: Training curves showing accuracy and loss across training phases for the four model variants

5. Discussion & Limitations

5.1. Performance Analysis

As shown in Table 1, ResNet50V2 with Preprocessing achieved the highest F1-score of 0.9443, outperforming all other configurations. Comparing the scenarios, preprocessing consistently improved results: EfficientNetB0's F1-score rose by 2.79% (from 0.8610 to 0.8889), and ResNet50V2's F1-score increased by 2.82% (from 0.9161 to 0.9443). This demonstrates that Gaussian Blur and CLAHE help models extract cleaner features by removing high-frequency noise and boosting contrast around surface spots.

5.2. Challenges

To understand specific errors, the confusion matrices for all four variants are displayed in Fig. 5.

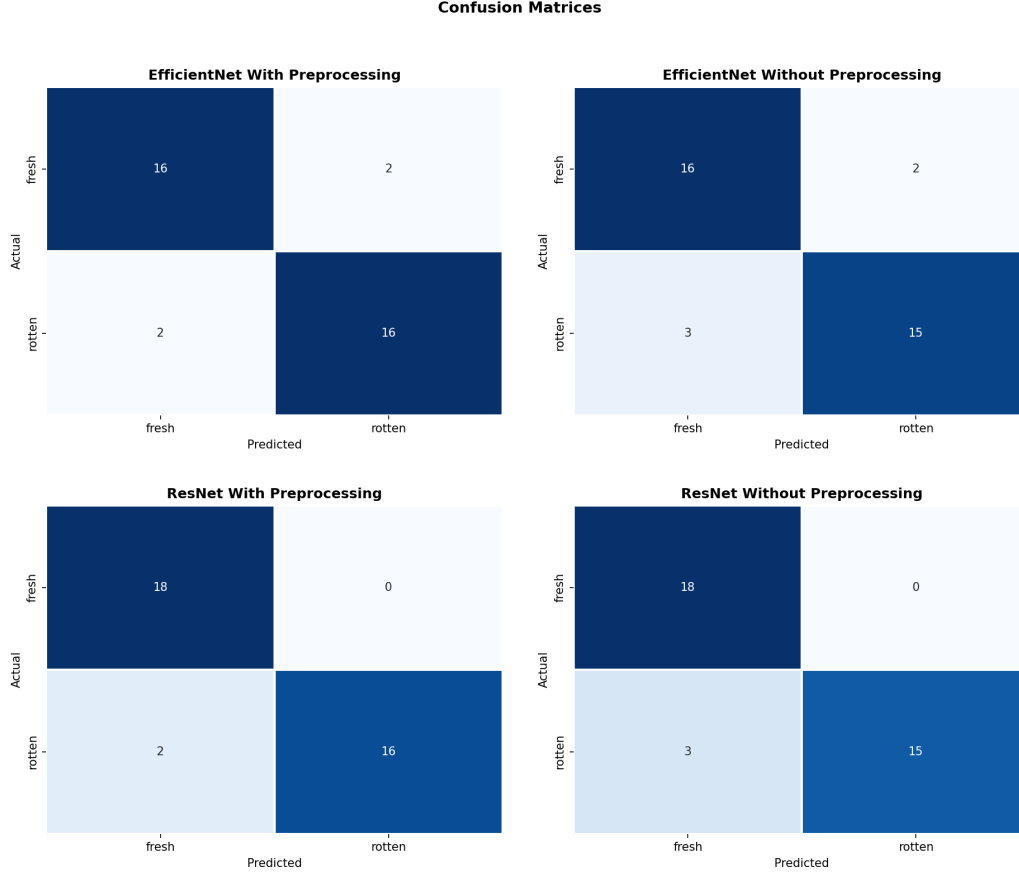


Figure 5: Confusion matrices for the four model variants

The most common error across all models was misclassifying rotten fruit as fresh (False Positives). This occurs because partially spoiled fruit often has large areas that still look completely fresh. If the model focuses on these fresh areas, it predicts the fruit as fresh. Background textures and lighting changes can also cover up small rotten spots. Applying CLAHE helped reduce this error by increasing the contrast of dark spots and mold, making it easier for the models to spot early signs of rot.

5.3. Trade-offs

There is a clear trade-off between model size and classification accuracy. EfficientNetB0 is extremely lightweight and fast, making it highly suitable for deployment on low-power edge devices. However, it achieved a lower F1-score compared to ResNet50V2. ResNet50V2, although computationally heavier and possessing more trainable parameters, achieved superior accuracy and recorded zero False Negatives on the test set.

6. Conclusion & Future Work

6.1. Conclusion

This project successfully developed a deep learning system to classify fruit freshness into fresh and rotten categories. We compared two model architectures (EfficientNetB0 and ResNet50V2) under two conditions (with and without a preprocessing pipeline of Gaussian Blur and CLAHE). The models were evaluated on a 36-image test set using Test Time Augmentation (TTA).

Our main findings are:

1. **Impact of Preprocessing:** Image preprocessing consistently improved performance. Gaussian Blur successfully removed background noise, while CLAHE enhanced details like dark spots and mold.
2. **Model Comparison:** ResNet50V2 outperformed EfficientNetB0 in both scenarios. Its residual connections and deeper custom head proved more capable of learning detailed textures and subtle color changes on the fruit surface.
3. **Best Configuration:** The ResNet50V2 model with preprocessing was the best-performing variant, achieving an F1-score of 0.9443 and misclassifying zero fresh images.

6.2. Future Work

To further enhance and scale this classification system, future research will focus on three key areas:

1. **Dataset Expansion:** Expanding the dataset size and variety of fruits to improve the model's generalization capabilities across different commodities.
2. **Alternative Architectures:** Testing lightweight architectures such as MobileNetV3 for mobile and edge deployment, as well as Vision Transformers (ViTs) for higher-resolution visual parsing.
3. **Real-time and Application Integration:** Deploying the model into real-time quality control systems (e.g., conveyor belt cameras) and integrating it into mobile applications for consumers and retailers.

7. Appendix

7.1. Screenshots

Examples of prediction results, showing the ground truth, model prediction, and confidence level, are displayed in Fig. 6 (for EfficientNetB0 with Preprocessing) and Fig. 7 (for ResNet50V2 with Preprocessing).

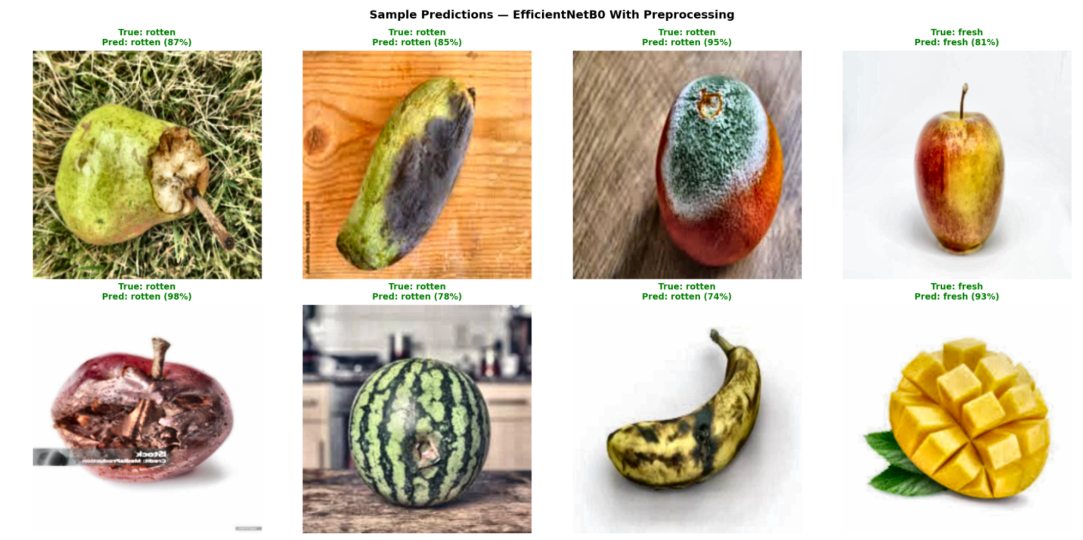


Figure 6: Example prediction results from the EfficientNetB0 model with preprocessing



Figure 7: Example prediction results from the ResNet50V2 model with preprocessing

7.2. Code Snippets

The core enhancement pipeline utilizing OpenCV for Gaussian Blur and CLAHE in the LAB color space is implemented as follows:

```
def _enhance(img: np.ndarray) -> np.ndarray:
    img = img.astype(np.uint8)
    # 1. Gaussian Blur
    img = cv2.GaussianBlur(img, (3, 3), 0.5)
    # 2. CLAHE on L channel (LAB space)
    lab = cv2.cvtColor(img, cv2.COLOR_RGB2LAB)
```

```
clahe = cv2.createCLAHE(clipLimit=2.5, tileGridSize=(8, 8))
L_channel_uint8 = lab[:, :, 0].astype(np.uint8)
lab[:, :, 0] = clahe.apply(L_channel_uint8)
return cv2.cvtColor(lab, cv2.COLOR_LAB2RGB)
```

7.3. Additional Attachment

- **Demo Video:** [Google Drive Folder](#)

References

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